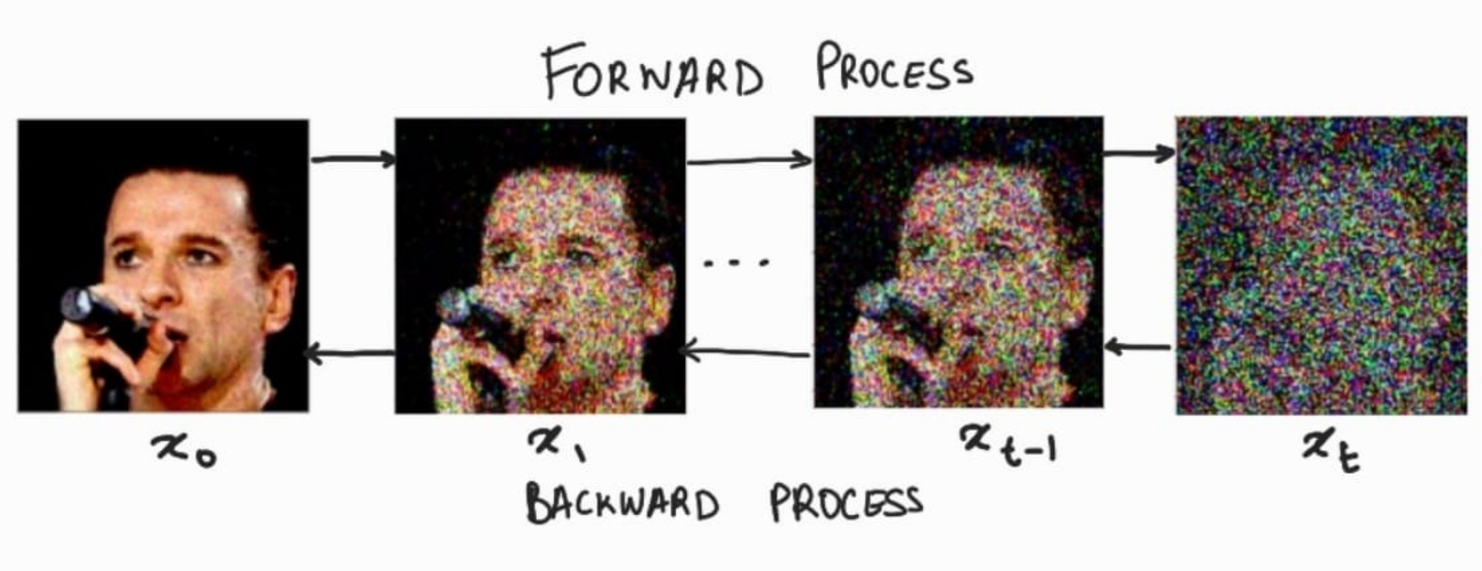
Entropy production



Macroscopic View (Shannon / Gibbs): Represents the total uncertainty or spread of the entire ensemble

$$S(t) = -\int p(x_t, t) ln p(x_t, t) dx$$

Microscopic View (Surprisal): Represents the uncertainty of a **single image** (a point x_t in high-dimensional space) within the distribution

$$s(x_t, t) = -\, ln p(x_t, t)$$

The ensemble average of individual surprisals perfectly recovers the global Shannon entropy.

$$\langle s(x, t) \rangle = S(t)$$

For a single image moving through the diffusion process from $t = 0$ to $t = T$, the total entropy produced is the sum of two distinct contributions:

System Entropy Change (Δs_{sys}): The net change in the image's "unexpectedness" (surprisal).

$$\Delta s_{sys} = -\ln p(X_T, T) + \ln p(X_0, 0)$$

Environmental Entropy Change (Δs_{env}): Information dissipated into the noise bath.

$$\Delta s_{env} = \frac{1}{D} \int_0^T f(X_t) dX_t$$

By expressing the Langevin constants in terms of the noise schedule $\beta(t)$ (where $k = \frac{1}{2}\beta(t)$ and $2D = \beta(t)$), we obtain:

Environmental Component (Dissipation): For an Ornstein-Uhlenbeck process, the environmental entropy simplifies to the change in the image's "distance" from the origin:

$$\Delta s_{env} = \frac{\frac{1}{2}\beta(t)(X_0^2 - X_T^2)}{\beta(t)} = \frac{X_0^2 - X_T^2}{2}$$

System Component (Surprisal Change): Calculated using the Gaussian distribution $p(x,t) = \mathcal{N}(\mu(t), \sigma^2(t))$

$$\Delta s_{sys} = \ln\left(\frac{p(X_0,0)}{p(X_T,T)}\right)$$

where $\mu(t) = x_0 e^{-kt}$ and $\sigma^2(t) = \frac{D}{\lambda}(1 - e^{-2\lambda t})$ are the analytical moments